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SOLUTION -: Practice Project Devops

```
import sys
def parse_log_file(log_file):
  logs = []
  with open(log_file, 'r') as file:
    for line in file:
      parts = line.split()

      if len(parts) < 5:
        continue

      ip = parts[0]
      endpoint = parts[2]
      try:
        response_time = float(parts[4])
      except ValueError:
        print(f"Warning: Skipping line due to invalid response time: {line.strip()}")
        continue

      logs.append({'ip': ip, 'endpoint': endpoint, 'response_time': response_time})
  return logs

def avg_response(log_file, endpoint):
  logs = parse_log_file(log_file)
  total_time = 0
  count = 0
```

```
    for log in logs:
        if log['endpoint'] == endpoint:
            total_time += log['response_time']
            count += 1
    return total_time / count if count > 0 else 0

def total_request_from_ip(log_file, ip):
    logs = parse_log_file(log_file)
    return sum(1 for log in logs if log['ip'] == ip)

def req_per_ip_per_endpoint(log_file, ip):
    logs = parse_log_file(log_file)
    endpoint_counts = {}
    for log in logs:
        if log['ip'] == ip:
            endpoint_counts[log['endpoint']] = endpoint_counts.get(log['endpoint'], 0) + 1
    return endpoint_counts

def req_ip_particular_ep(log_file, ip, endpoint):
    logs = parse_log_file(log_file)
    return sum(1 for log in logs if log['ip'] == ip and log['endpoint'] == endpoint)

if __name__ == "__main__":
    log_file = sys.argv[1]
    ip = sys.argv[2]
    endpoint = sys.argv[3]
    print("avg_response -", avg_response(log_file, endpoint))
    print("total_request_from_ip -", total_request_from_ip(log_file, ip))
```

```
    print("req_per_ip_per_endpoint -", req_per_ip_per_endpoint(log_file, ip))
    print("req_ip_particular_ep -", req_ip_particular_ep(log_file, ip, endpoint))
```

CODE and OUTPUT window Screen Shot is attached -:

test.py 4 | file.log | log_solution.py | dockerfile

Project > log_solution.py > req_per_ip_per_endpoint

```python
import sys

def parse_log_file(log_file):
    logs = []
    with open(log_file, 'r') as file:
        for line in file:
            parts = line.split()

            if len(parts) < 5:
                continue

            ip = parts[0]
            endpoint = parts[2]
            try:
                response_time = float(parts[4])
            except ValueError:
                print(f"Warning: Skipping line due to invalid response time: {line.strip()}")
                continue

            logs.append({'ip': ip, 'endpoint': endpoint, 'response_time': response_time})
    return logs

def avg_response(log_file, endpoint):
    logs = parse_log_file(log_file)
    total_time = 0
    count = 0
    for log in logs:
        if log['endpoint'] == endpoint:
            total_time += log['response_time']
            count += 1
    return total_time / count if count > 0 else 0

def total_request_from_ip(log_file, ip):
    logs = parse_log_file(log_file)
    return sum(1 for log in logs if log['ip'] == ip)

def req_per_ip_per_endpoint(ip,log_file):
    logs = parse_log_file(log_file)
    endpoint_counts = {}
```

test.py 4 | file.log | log_solution.py | dockerfile

Project > log_solution.py > req_per_ip_per_endpoint

```python
def req_per_ip_per_endpoint(ip,log_file):
    logs = parse_log_file(log_file)
    endpoint_counts = {}
    for log in logs:
        if log['ip'] == ip:
            endpoint_counts[log['endpoint']] = endpoint_counts.get(log['endpoint'], 0) + 1
    return endpoint_counts

def req_ip_particular_ep(log_file, ip, endpoint):
    logs = parse_log_file(log_file)
    return sum(1 for log in logs if log['ip'] == ip and log['endpoint'] == endpoint)

if __name__ == "__main__":
    log_file = sys.argv[1]
    ip = sys.argv[2]
    endpoint = sys.argv[3]

    print("avg_response -", avg_response(log_file, endpoint))
    print("total_request_from_ip -", total_request_from_ip(log_file, ip))
    print("req_per_ip_per_endpoint -", req_per_ip_per_endpoint(ip,log_file))
    print("req_ip_particular_ep -", req_ip_particular_ep(log_file, ip, endpoint))
```

PROBLEMS 4 | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS

```
(base) Meenakshis-MacBook-Pro:Project meenakshi$ python3 log_solution.py file.log 139.23.98.21 /abc
avg_response - 0.31666666666666665
total_request_from_ip - 30
req_per_ip_per_endpoint - {'/abc': 20, '/jkl': 10}
req_ip_particular_ep - 20
(base) Meenakshis-MacBook-Pro:Project meenakshi$
```

bash

Python Project